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SUMMARY

- ◆ Provides the historical and methodological grounding for understanding participatory design as a methodology
- ◆ Describes its research designs, methods, criteria, and limitations
- ◆ Provides guidance for applying it to technical communication research

The Methodology of Participatory Design

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INTRODUCTION

Technical communicators have begun writing quite a bit about participatory design, sometimes with a fervor that rivals that with which we used to write about T-units or think-aloud protocols. The terms *participatory design* and *user-centered design* are being broadly applied in the philosophical and pedagogical work of technical communication (Blythe 2001; Henry 1998; Johnson 1998; Salvo 2001; Spinuzzi 2003); methods associated with those terms are being applied in technical communication research (Mirel 1988, 2003; Smart 2003; Smart and Whiting 2002; Smart, Whiting, and DeTienne 2002; Spinuzzi 2002a, 2002c, in press; Wixon and Ramey 1996); and prototypes in particular are often presented as a vital part of iterative usability (see, for example, Barnum 2002, Chapter 4; Smart and Whiting 2002). But that breadth of application has often come at the price of imprecision. It's hard to find a good *methodological* explanation of participatory design.

That lack of a strong methodological explanation is not just technical communication's problem, though. Participatory design is often discussed in human-computer interaction, computer-supported cooperative work, and related fields as a *research orientation* or even a *field* (see Muller 2002, p. 1,052) rather than a methodology. The distinction may be important in principle, but in practice, it has become an escape hatch that allows practitioners to label their work "participatory design" without being accountable to established, grounded precedent.

By looking at that established precedent, I argue, we *can* define participatory design as a methodology, even if it's a loose one. And I believe it's time we did: Without such a definition, we can't hold ourselves accountable to participatory design or build on a coherent body of knowledge. Consequently, we have trouble applying participatory design rigorously to our technical communication projects, and we tend to think of participatory design as an approach

to design rather than a rigorous research methodology.

In this article, I discuss participatory design as a research methodology, characterizing it as a way to understand *knowledge by doing*: the traditional, tacit, and often invisible (in the sense of Nardi and Engeström 1999; Muller 1999) ways that people perform their everyday activities and how those activities might be shaped productively. I first define and describe *what* participatory design research is. I describe participatory design research in terms of its paradigm, methodology, research design, and methods. With this definition and description as a framework, I next discuss *why* we should pursue participatory design studies. In this section, I discuss the benefits of knowledge by doing and provide evaluative criteria to use as guidelines for creating and assessing participatory design research. Finally, I explore the implications of understanding participatory design as a research methodology, and I discuss some practical applications.

WHAT IS PARTICIPATORY DESIGN RESEARCH?

Participatory design *is* research. Although it has sometimes been seen as a design approach characterized by user involvement (Johnson 1998), participatory design has its own highly articulated methodological orientation, methods, and techniques, just as does participatory action research, the approach on which it is based (Glesne 1998). Implementations of participatory design do vary in their attention to rigor and validity (Spinuzzi in press), but they all reflect a commitment to sustained, methodical investigation according to grounded methodological principles, as we'll see below.

Participatory design is rather different from most research conducted by technical communicators, though it

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turns out to be a good match for the work we do. As the name implies, the approach is just as much about *design*—producing artifacts, systems, work organizations, and practical or tacit knowledge—as it is about *research*. In this methodology, design *is* research. That is, although participatory design draws on various research methods (such as ethnographic observations, interviews, analysis of artifacts, and sometimes protocol analysis), these methods are always used to iteratively construct the emerging design, which itself simultaneously constitutes and elicits the research results as co-interpreted by the designer-researchers and the participants who will use the design.

Like member checks in ethnographic research, participatory design's many methods ensure that participants' interpretations are taken into account in the research. Unlike member checks, however, these methods are shot through the entire research project; the goal is not just to empirically understand the activity, but also to simultaneously envision, shape, and transcend it in ways the workers find to be positive. In participatory design, participants' cointerpretation of the research is not just confirmatory but an essential part of the process.

Participatory design started in Scandinavia through a partnership between academics and trade unions. Since that time it has worked its way across the Atlantic, becoming an important approach for researchers interested in human-computer interaction, computer-supported cooperative work, and related fields. From there, it has begun to influence writing studies, particularly through technical communication as well as computers and composition (for example, Sullivan and Porter 1997; Johnson 1998; see Spinuzzi 2002b for an overview).

Participatory design has undergone many changes—for instance, later variations have moved away from the Marxist underpinnings of the earlier work—but its core has remained more or less constant. It attempts to examine the tacit, invisible aspects of human activity; assumes that these aspects can be productively and ethically examined through design partnerships with participants, partnerships in which researcher-designers and participants cooperatively design artifacts, workflow, and work environments; and argues that this partnership must be conducted iteratively so that researcher-designers and participants can develop and refine their understanding of the activity. The result of the research typically consists of designed artifacts, work arrangements, or work environments.

As Pelle Ehn suggests, participatory design attempts to steer a course “between tradition and transcendence” (1989, p. 28)—that is, between participants' tacit knowledge and researchers' more abstract, analytical knowledge. The developers of participatory design believed that politically and ethically, the two types of knowledge must be bridged, with each being valued by all involved in the

research. That's especially true in studies of workers, for which participatory design was initially designed, but also in studies of end users and students.

History

Participatory design originated in Scandinavia in the 1970s and 1980s. This early Scandinavian work was motivated by a Marxist commitment to democratically empowering workers and fostering democracy in the workplace. This avowedly political research aimed to form partnerships with labor unions that would allow workers to determine the shape and scope of new technologies introduced into the workplace. Up to that point, labor unions had little experience with computer technologies and had been forced to accept systems developed by management, systems that represented a sharp break from workers' traditional ways of working; exerted a greater and greater control over increasingly fine details of their work; and automated large swathes of the workflow, putting people out of work (see Ehn, 1990; Zuboff, 1989).

Since they did not know how to design computer technologies themselves, workers were put into the position of accepting these disempowering technologies or simply rejecting them. Some Scandinavian researchers set out to develop a third way, an approach that provided a set of “language games” (Ehn and Kyng 1991, pp. 176–177) that would allow software developers and workers to collaboratively develop and refine new technologies—allowing workers to retain control over their work.

These researchers turned to *action research*, in which ethnographic methods are linked to positive change for the research participants (see Glesne 1998 for an overview). Clement and van den Besselaar explain that

Unlike conventional research, which is directed primarily at producing results of interest to those beyond the immediate research site, an essential goal of action research is to achieve practical or political improvements in the participants' lives (e.g., less routine work, greater autonomy, more effective tools). The researcher becomes directly involved in the ongoing work and feeds results back to the participants. (1993, p. 33)

Action research involves alternating between *practical work* to support changes (such as design activities) on one hand, and *systematic data collection and analysis* on the other hand (p. 33; see also Bertelsen 2000). In early participatory design studies, workers tried to describe computer systems that could automate work while still valuing their craft skills and upholding their autonomy. But because workers had no experience in systems design, they could not begin to speculate on how to build such a system. So in the UTOPIA project, researchers joined with

a workers' union to experiment with a range of research techniques, including mockups and other low-fidelity prototypes, future workshops, and organizational toolkits (Bødker and colleagues 1987). Although the project failed to produce a working system, it did produce a design approach and a range of techniques for participatory design work. Based on UTOPIA and other projects that came after it, the Scandinavians issued the "Scandinavian challenge": to develop and use design approaches that encourage industrial democracy (Bjerknes, Ehn, and Kyng 1987). This call resulted in many approaches and techniques under the umbrella of participatory design, such as CARD, PICTIVE, cooperative interactive storyboard prototyping, and contextual design (see Muller, Wildman, and White 1993 for an exhaustive taxonomy). Some of these, such as contextual design, have become complex enough to be categorized on their own and, arguably, differentiated enough that they are no longer participatory design in the strictest sense of the term (see Spinuzzi 2002c). In the United States, because of relatively weak labor unions and a focus on functionality rather than workplace democracy (see Spinuzzi in press), participatory design has tended to be implemented through *nonintrusive* methods: workplace microethnographies rather than walkthroughs and workshops (Blomberg and colleagues 1993; Blomberg, Suchman, and Trigg 1997), small-scale card-matching exercises rather than large-scale organizational games (Muller and Carr 1996), and one-on-one prototyping sessions that focus on confirming developed ideas rather than group prototyping sessions that emphasize exploration (Beyer and Holtzblatt 1998; Spinuzzi 2005). But the basic methodological principles of participatory design remained. What distinguishes participatory design from related approaches such as user-centered design is that the latter supposes only that the research and design work is done on *behalf* of the users; in participatory design, this work must be done *with* the users (Iivari 2004).

Defining users' knowledge

Participatory design's object of study is the *tacit knowledge* developed and used by those who work with technologies. It's important to understand this focus because tacit knowledge, which is typically difficult to formalize and describe, has tended to be ignored by the theory of cognition that has tended to dominate human-computer interaction: information processing cognitive science (Winograd and Flores 1986; Nardi 1996; Nardi and Engeström 1999).

In practice, this theory tends to lead to a rationalist approach to design, which generally assumes that there is one best way to perform any activity—an assumption it shares with Taylorism. This rationalist approach was something to which early participatory designers reacted strongly. They were heavily influenced by Marxist critiques

of Taylorism, such as Harry Braverman's argument that Taylorism seeks to effect managerial control through "*the dictation to the worker of the precise manner in which work is to be performed*" (1974, p. 90, emphasis his). That is, rather than allowing workers to determine how to accomplish their tasks—and develop their own tacit craft skills and knowledge not possessed by management—the Taylorist manager examines the work, then breaks it into discrete, formal tasks that can be optimized, regulated, and taught to new workers. All discretion and all decisions are taken away from the workers. Knowledge is made explicit, formalized, and regulated; workers' craft traditions are judged inferior. (See Muller 1997, 1999 for discussions of this tendency in U.S. corporations and a response from the perspective of participatory design.)

Participatory design opposes this notion of knowledge on both political and theoretical grounds. Politically, this notion of knowledge as wholly consisting of optimized tasks spells the death of workplace democracy: if it is accepted, workers cannot have a say in their own work because only trained researchers can determine the best way of performing that work. Theoretically, participatory design is founded on constructivism, a theory that explicitly resists the notion that knowledge can be completely formalized and classified. (For overviews of the constructivist argument in writing studies, see Mirel 1998; Spinuzzi 2003).

Knowledge is situated in a complex of artifacts, practices, and interactions; it is essentially interpretive, and therefore it cannot be decontextualized and broken into discrete tasks, nor totally described and optimized. In the constructivist view, participants' knowledge is valorized rather than deprecated, and their perspectives therefore become invaluable when researching their activity and designing new ways to enact that activity. "Knowing and learning," as Barbara Mirel says, "take place in a dynamic system of people, practices, artifacts, communities, and institutional practices" (1998, p. 13).

When we think of *knowledge*, we often think of explicit forms of knowledge: things that are written down, defined, categorized, systematized, or quantified. But to understand knowledge-making in participatory design, we have to understand that much knowledge tends to be *tacit*. Tacit knowledge is implicit rather than explicit, holistic rather than bounded and systematized; it is what people know without being able to articulate. As Ehn argues, participatory design takes a Heideggerian approach to knowledge in which "*the fundamental difference between involved, practical understanding and detached theoretical reflection is stressed*" (1989, p. 28). This pragmatic approach involves alternating between the two by discovering tacit knowledge, then critically reflecting on it.

Since practical tacit knowledge was a main goal of early participatory design research, researchers adopted

the *tool perspective*, the idea that “computer support is designed as a collection of tools for the skilled worker to use. The tool perspective takes the work process as its origin rather than data or information flow. This means: *not* detailed analysis, description, and formalization of qualifications, but the development of professional education based on the skills of professionals; *not* information flow analysis and systems description but specification of tools” (Bødker and colleagues 1987, p. 261).

The tool perspective allowed researchers to recognize and leverage the workers’ craft knowledge, allowing them to develop new tools that would support rather than disrupt that work: “a tool is developed as an extension of the accumulated knowledge of tools and materials within a domain” (Bødker and colleagues 1987, p. 261; see also Ehn 1989, pp. 339–40). In contrast to design approaches favored by management that served Tayloristic goals (deskilling, work intensification), the tool perspective involved building “computer-based tools by which the craftsman can still apply and develop original skills” (p. 261; see Ehn 1989, p. 34; Ehn and Kyng 1987, pp. 34–38).

This tacit or craft knowledge is linked to *metis*: “Metis, or what is also called cunning intelligence, is the ability to act quickly, effectively, and prudently within ever-changing contexts” (Johnson 1998, p. 53). These ever-changing contexts are what Mirel points to when she talks about complex tasks (1998, 2004). In participatory design, tacit knowledge is not only explored, it is in many cases made material, as we saw with the tool perspective that participatory designers adopted. Workers find unconventional ways to use the tools that have been supplied to them, learn how to construct their own ad hoc tools (Spinuzzi 2003), and—if they are allowed the time and freedom to do so—eventually stabilize new tools and the ways they interact with them.

One goal of participatory design is to preserve tacit knowledge so that technologies can *fit into* the existing web of tacit knowledge, workflow, and work tools, rather than doing away with them. In contrast to rationalist studies that assume workers’ tasks can be broken down into their components, formalized, and made more efficient, participatory design assumes that tacit knowledge cannot be completely formalized; the task-and-efficiency orientation typical in many user-centered design methods such as GOMS (Card, Moran, and Newell 1983; Muller 1999) and usability testing (Barnum 2002; Rubin 1994) can actually get in the way of the holistic activity.

Certainly, some tacit knowledge can be made explicit and formalized, but

attempts at explication of such tacit knowledge must always be incomplete. The knowledge is too layered and subtle to be fully articulated. That is why action-

centered skill has always been learned through experience (on-the-job training, apprenticeships, sports practice, and so forth). Actions work better than words when it comes to learning and communicating these skills. (Zuboff 1988, p. 188)

So tacit knowledge often remains *invisible*: since it is not made systematic or quantifiable, it passes unnoticed and often undervalued. (See Nardi and Engeström 1999 for a collection of essays on this theme.) In particular, low-level workers are often not valued by management because their skills are invisible: the complexity, difficulty, and interconnectedness of their work are not recognized. One example is Blomberg, Suchman, and Trigg’s (1994) study in which document analysts (temporary workers who coded legal documents) were found to perform complex interpretive work. The attorneys who employed these workers did not recognize the work as being complex or interpretive, and consequently planned to outsource the work to lower paid workers in another country. Like others working in the participatory design tradition, Blomberg, Suchman, and Trigg attempted to demonstrate to management the tacit knowledge that workers brought to the activity, knowledge that had remained invisible up to that point, yet was vital to the continued success of the activity.

Describing users’ knowledge

Since users’ tacit knowledge is highly valued, participatory design focuses on exploring that tacit knowledge and taking it into account when building new systems. This task is accomplished with a strong political or ethical orientation: users’ knowledge is described so that it can be used to design new tools and workflows that empower the users. (What is meant by *empowerment* is sometimes different in the different strands of participatory design.) In this section, I describe the paradigm that underpins participatory design, its methodology, research design, and methods.

Paradigm Participatory design’s paradigm is *constructivist* in Mirel’s sense (1998). That is, it sees knowledge-making as occurring through the interaction among people, practices, and artifacts—knowledge doesn’t just reside in the head; it’s a condition of a certain context. One of the most distinct and influential notions of participatory design is that of the *language game* (Ehn 1989, p. 17): bridging the worlds of researcher-designers and users by finding a common “language” or mode of interaction with which both parties feel comfortable.

Methodology Participatory design’s methodology is derived from participatory action research or, as Ehn calls it, “practice research”: “Practical interventionistic investigations (as opposed to gathering of data) and parallel theo-

retical reflection (as opposed to detached theoretical reflections *a posteriori*)” (Ehn 1989, p. 13). As discussed above, this activist brand of research has an explicit political-ethical orientation: to empower workers to take control over their work. Unlike Donald Norman—who argues that the designer should be a *dictator* (Grossman 2002)—participatory designers see themselves as *facilitators* who attempt to empower users in making their own decisions (Clement 1994).

To achieve that goal, participatory design emphasizes co-research and co-design: researcher-designers must come to conclusions in conjunction with users. So participatory design involves redesigning workplaces and work organization as well as work tools. And it is iterative, allowing workers and researchers to critically examine the impacts of these incremental redesigns in progress.

Research design Participatory design is still developing and consequently its research design tends to be quite flexible. For instance, the early Scandinavian work tended to rely on union-sponsored workshops and games involving heavy direct interaction between designers and users, while later work in the U.S. has tended to supplement targeted interaction with less intrusive methods such as observation and artifact analysis. But three basic stages are present in almost all participatory design research:

◆ **Stage 1: Initial exploration of work**

In this stage, designers meet the users and familiarize themselves with the ways in which the users work together. This exploration includes the technologies used, but also includes workflow and work procedures, routines, teamwork, and other aspects of the work.

◆ **Stage 2: Discovery processes**

In this stage, designers and users employ various techniques to understand and prioritize work organization and envision the future workplace. This stage allows designers and users to clarify the users’ goals and values and to agree on the desired outcome of the project. This stage is often conducted on site or in a conference room, and usually involves several users.

◆ **Stage 3: Prototyping**

In this stage, designers and users iteratively shape technological artifacts to fit into the workplace envisioned in Stage 2. Prototyping can be conducted on site or in a lab; involves one or more users; and can be conducted on-the-job if the prototype is a working prototype.

The stages can be (and usually should be) iterated several times. Together, they provide an iterative co-exploration by designers and users.

Methods Methods are grouped by stage.

◆ **Stage 1: Initial exploration of work**

Since initial exploration tends to involve examining technology use on site, Stage 1 draws from ethnographic methods such as observations, interviews, walkthroughs and organizational visits, and examinations of artifacts. This stage is typically conducted on site, during the normal work day. In the earlier Scandinavian iterations, this initial exploration tended to be highly interactive and intrusive: the researchers generally aligned themselves with relatively powerful workers’ unions that believed in the projects and could insist on the sorts of disruptions caused by walkthroughs and organizational visits (see Bødker, Grønbaek, and Kyng 1993 for an overview).

In North America, unions were much weaker and workers were not in a position to force participation, nor were they terribly interested in such projects. So researchers turned to less intrusive ethnographic and ethnomethodological techniques such as observations and interviews (see Wall and Mosher 1994 for an overview). Although the methods draw from ethnography, they are oriented toward design as well as description, so they tend to be focused and enacted differently, with more interaction in mind (see Beyer and Holtzblatt 1998 for one example, and Spinuzzi 2002c and in press for critiques). Much of that interaction takes place during the second stage, in discovery processes.

◆ **Stage 2: Discovery processes**

Stage 2 is where researchers and users interact most heavily, and it also typically involves group interactions. Again, discovery processes tended to be more interactive and intrusive in the earlier Scandinavian iterations than in the later North American iterations, but in all implementations they are more interactive than traditional ethnographies. Because of participatory design’s orientation toward design, the goal is to cooperatively make meaning out of the work rather than to simply describe it. Methods used during this stage include organizational games (Bødker, Grønbaek, and Kyng 1993, pp. 166–167), role-playing games (Iacucci, Kuutti and Ranta 2000), organizational toolkits (Tudor, Muller, and Dayton 1993; Ehn and Sjögren 1991; Bødker and colleagues 1987), future workshops (Bødker, Grønbaek, and Kyng 1993, p. 164; Bertelsen 1996), storyboarding (Madsen and Aiken 1993), and workflow models and interpretation sessions (Beyer and Holtzblatt 1998).

◆ **Stage 3: Prototyping**

Finally, this stage involves a variety of techniques for iteratively shaping artifacts. These techniques include mockups (Ehn 1989; Ehn and Kyng 1991; Bødker

and colleagues 1987), paper prototyping (Novick 2000), cooperative prototyping (Bødker and Grønbaek 1991; Grønbaek and Mogensen 1994); and PICTIVE (Muller 1991b, 1993), among many others.

Finally, and just as importantly, results are disseminated in forms that users can understand and share—a continuation of the “language games” that allow researchers and users to collaborate, and a way to continue to support the empowerment and participation of users. The tone for this dissemination was set early on, in the UTOPIA project: results were discussed in everyday language in a union publication called *Graffiti* (Ehn 1989, pp. 350–352). Another example is contextual design’s practice of “walking” through affinity diagrams and consolidated models with participants and of providing a room with diagrams and prototypes posted on the walls so that workers, managers, engineers, marketing people, and customers can see the state of the project in progress (Beyer and Holtzblatt 1998, chapter 10).

CRITICALLY EXAMINING PARTICIPATORY DESIGN STUDIES

Despite its advantages, participatory design has some rather sharp limitations as well as some criteria for success that are not immediately obvious. Below, I review some of the limitations of participatory design and discuss criteria for evaluating participatory design studies.

Limitations of participatory design

Participatory design has strengths, but as with other research approaches, those strengths come with tradeoffs.

Limitations of methodology Since participatory design aims to ground changes in traditional craft skills as a way of empowering workers, some argue that participatory design does not lend itself to radical change of the sort that sometimes must characterize new systems (Beyer and Holtzblatt 1998). In fact, participatory designers have been cautioned to think of their work as “evolution, not revolution” (Sumner and Stolze 1997). This gradualist tendency can lead to tunnel vision, in which particular stakeholders are served while others are left to fend for themselves (Bjerknes and Bratteteig 1995; Bødker 1996). In response, some participatory designers have worked to bring in new accounts of stakeholders that can support more complex projects (Bødker 1996; Muller 2003).

Another limitation is that some strains of participatory design—particularly later work that emphasizes functional empowerment over democratic empowerment, such as cooperative prototyping (Bødker and Grønbaek 1991)—have a tendency to focus too narrowly on artifacts rather than overall workflow, presuming that fine-tuning the artifact will necessarily result in empowering changes to the overall work activity (Spinuzzi 2002c). Finally, as participatory

design has migrated across socioeconomic borders, from Scandinavia to North America, researchers have had difficulty maintaining its methodological tenets, particularly its focus on democratic empowerment (Muller 1991a; Spinuzzi 2002b, in press).

Limitations of method If more rigorous methods can be described as “measure twice, cut once,” participatory design methods can be described as “explore, approximate, then refine.” This essentially dissimilar methodological orientation—related to action research’s juggling act between the traditional researcher’s role of collecting and analyzing data *versus* the activist’s role of initiating and sustaining significant change at the research site—tends to alter how researcher-designers apply established methods. For instance, participatory design researchers often draw on ethnographic methods to develop knowledge about the participants’ work, tools, and craft traditions. But these researchers, who often come from backgrounds in systems design, human-computer interaction, or technical communication, tend to apply these methods quite loosely in the eyes of trained ethnographers.

Diana Forsythe, for instance, scathingly critiques these applications as “do-it-yourself ethnography” and complains that “superficial social research may confer the illusion of increased understanding when in fact no such understanding has been achieved.” She specifically takes to task a contextual design project “in which brief exercises in shadowing, observation, and interviewing have been undertaken from a common sense stance without engaging the questions that define ethnography as anthropologists understand it,” and warns that “such an exercise can result in a cognitive hall of mirrors. Without addressing basic issues such as the problem of perspective, researchers have no way of knowing whether they have really understood anything of their informants’ world view or have simply projected and then ‘discovered’ their own assumptions in the data” (1999, p. 136; see also Cooper and colleagues 1995; Nyce and Lowgren, 1995).

Forsythe’s critique is valid if the aim of research is to extract knowledge in the mode of traditional research, pulling the data into another domain where it can be abstracted, analyzed, and used apart from the site. But participatory design research, properly done, continually brings the analysis back to the domain and shares it with the participants, who co-interpret it, co-analyze it, and co-design responses to it. That is, the traditional methods are—at least in the best examples—re-networked or reconfigured to meet the design orientation.

The “same” methods can be enacted differently and take rather different shapes as they are attached to different methodologies and paradigms. In this case, the resulting research and designs do give up traditional research rigor,

but they do so to gain reflexivity and agreement. (In the earlier, highly politicized Scandinavian work, that agreement took the shape of political representation; in later work, the focus shifted to ethical concerns in giving workers the tools needed to do their jobs, and agreement took the shape of consensus among representative users.) This tradeoff resembles “rolling” member checks.

For example, Muller (1999) describes using the participatory design technique CARD to study the work of telephone operators. CARD, he says, has less rigor and predictive power than more narrowly defined analytical techniques such as GOMS, but on the other hand it brings in benefits that are more important to participatory designers:

Its strengths lie in its ability to capture diverse information . . . , its openness to the disconfirmation of assumptions . . . , and its extensibility in the face of new information. Underlying all of these attributes is CARD's enfranchisement of multiple stakeholders with differing disciplines, perspectives, and positions. (p. 54; see also Bertelsen 2000)

Rigor is difficult to achieve because researchers cede considerable control to their participants and share a “design language” with those participants which must by its nature be imprecise. On the other hand, the proof is in the pudding, so to speak—the design artifact both *encapsulates* the research results (as the material trace left by the design efforts) and *elicits* them (both during design sessions and afterwards, as it is introduced into the environment to be used as a stable work artifact). Wall and Mosher demonstrate that the same design artifacts can be used as records of a field study; tools for analysis; communication tools for a language game in which researcher-designers and users participate; and focal artifacts for co-design and co-development (1994). Rigor becomes something different in participatory design research: a desirable goal, but subordinated to users' control and aims.

Practical limitations In addition to the methodological and methodical critiques is the practical one: participatory design research takes an enormous amount of time, resources, and institutional commitment to pull off. That institutional commitment in particular can be hard to come by. From the standpoint of a profit-oriented business, participatory design seems to provide little structure and no deadlines (Wood and Silver 1995, pp. 322–323). Researchers find that they have to cede considerable control to workers, who must be committed to the process and cannot be coerced. For example, Bertelsen (1996) ruefully recounts how some of his participants simply failed to show up for a future workshop, compromising the design

developed in the workshop. Finally, unlike ethnographic studies, participatory design studies typically require continuous critical participation by workers. Later participatory design variants such as contextual design (Beyer and Holtzblatt 1998) and customer partnering (Hackos, Hammar, and Elser 1997) have compromised by sharply limiting users' participation.

Evaluating participatory design

Participatory design is usually brought in at major turning points when work is to be automated and tools and workflows are to be changed. Since participatory design projects by definition involve design as well as research, the object of the research tends to be expressed in a *purpose statement* rather than a *research question*:

The purpose of this project . . . is to design a number of computer applications for [an organization] and to develop a long-term strategy for decentralizing development and maintenance. (Bodker, Gronbaek and Kyng 1993, p. 161)

The overall object of the project has been to contribute to the development of skill-enhancing tools for graphic workers. (Bodker and colleagues 1987, p. 254)

The work-oriented design project was originally conceived to explore bringing together the worlds of corporate research, product development, and specific work-sites . . . in an effort to design more useful new technologies. (Blomberg, Suchman, and Trigg 1997, p. 269)

In concert with these types of research statements, participatory design has developed criteria that are also oriented toward development. Participatory design is still a relatively young approach, and at present it is more of a movement or research orientation than a coherent methodology, so it hasn't developed evaluative criteria to the same level that, say, experimental studies have. But we can draw nascent criteria from the methodological principles discussed earlier. They are often difficult to meet. As Blomberg and Henderson (1990) illustrate, it's easy to produce a study that looks like participatory design but that fails at all three of the criteria listed here. Participatory design projects, despite their ceding of power and analysis to users, still must rigorously apply these criteria to have internal integrity.

Criterion #1: Quality of life for workers Most participatory designers would point to this criterion as the most important one. Participatory design is meant to improve workers' quality of life both in terms of demo-

cratic empowerment (that is, workers' control over their own work organization, tools, and processes) and functional empowerment (that is, workers' ability to perform their given tasks with ease; see Blomberg, Suchman, and Trigg 1997; Spinuzzi and colleagues 2003; Spinuzzi in press). In a participatory design study, workers critically reflect on their own practices, work organization, and tools.

In the earlier Scandinavian iterations, this critical reflection usually involved examining ways that workers could better control the terms of their work; in later U.S.-based iterations, critical reflection turned to an examination of tacit knowledge to more effectively meet the goals of the work. Either way, this methodological principle translates into an exploration of tacit knowledge, invisible work, and unstated individual and organizational goals.

To meet this criterion, participatory design studies strive for

- ◆ **Reflexivity and agreement** between researchers and users. The two groups interact closely through interviews, focus groups, workshops, organizational games, prototyping sessions, and other techniques to continually reassess the activity under investigation and to synchronize their interpretations.
- ◆ **Codetermination** of the project by researchers and users. Specific project criteria are codetermined by researchers and users during the project. This way, researchers do not take total ownership of the project; users are also able to shape the project to reflect their values, goals, and ends.

Criterion #2: Collaborative development Collaborative development is a key part of the effort to improve workers' quality of life. As noted earlier, users' work is often invisible and their knowledge is often tacit. Thus designers of information systems, educational Web sites, and documentation often assume that the work is simple, easily formalized, and (sometimes) easily automated. Collaborative development allows researcher-designers to avoid that trap by inviting participants to be co-researchers and co-developers. Doing so allows researcher-designers to elicit and explore the tacit knowledge and invisible practices that might otherwise have been lost, and simultaneously encourages workers to participate in their own empowerment.

In terms of a study criterion, this methodological principle translates into a requirement for mechanisms to ensure that data collection and analysis be done in conjunction with participants. In ethnographical terms, participatory design uses member checks—but in participatory design, the member checks are continuous since the project is co-owned and co-enacted by the participants. To meet this criterion, participatory design studies strive for

◆ **Involvement** The successful study will provide mechanisms for participation and produce verifiable changes based on them. Participatory design studies are not a “listening tour” in which researchers hear the concerns of users, then go away and design a solution; they are participatory top to bottom and must include verifiable, regular avenues for group interaction and definite routines for ensuring that users' concerns are methodically addressed in the resulting design.

◆ **Mechanisms for consensus/agreement and representation** In most cases, not every user can be involved in a participatory design study. For instance, if a participatory design study involved redesigning an interface used by 2000 workers, it's simply not practical or manageable to involve every worker in workshops and prototyping sessions. Instead, workers must be *represented* in the same way that politicians are elected to represent the interests and views of their constituencies.

In the earlier Scandinavian iterations of participatory design, representatives were assigned to projects by their unions, making them explicitly political representatives. In North America, however, unions hold considerably less power and no other ready-made mechanisms for political representation of workers exist; rather, workers are typically selected by management and are seen as *functionally* representative (that is, “average users”). In any case, users must be given the opportunity to be broadly represented in the study, and the representatives should have a way to settle disagreements or come to consensus.

Common language games such as contextual design's work diagrams and PICTIVE's pictures. To collaboratively develop solutions, users should be able to interact with researchers in a neutral “language” understood by both sides. It's not enough to *offer* such a language game; researchers must also confirm that users are comfortable with the language game, able to understand it, and able to use it both to critique solutions and to express their own solutions.

Common aims codetermined by researchers and users in advance. Near the beginning of the project, researchers and users should be able to settle on a list of common aims that represent the users' interests. That list must be flexible, as users will continue to critically evaluate their own aims.

Criterion #3. Iterative process But to enact collaborative development, researcher-designers and participants must follow an iterative process. Tacit knowledge and invisible practices are by their nature difficult to tease out.

A crude caricature of participatory design might involve gathering workers' comments on current practice and their responses to a prototype, but without sustained, iterative reflection on and use of a designed artifact, workers may not be able to comment critically or respond effectively (see Hackos, Hammar, and Elser 1997). Each change in a prototype tends to unearth other invisible work practices and other tacit knowledge.

In terms of a criterion for a study, this methodological principle translates into a requirement for a series of opportunities to sustain the continuous member check. To meet this criterion, participatory design studies strive for

Continual participation Users should be involved repeatedly or continually and offered mechanisms for co-design at multiple stages.

Revisiting stages Rarely is one sweep through the stages enough because the stages are designed to inspire critical reflection on the work and turn up tacit knowledge. So a successful participatory design project should be flexible enough to revisit stages repeatedly and cyclically.

Sustained reflection Finally, the continuous member check must go beyond simply reacting to the functionality of designs—a danger especially in the later stages of a project, when functioning prototypes take on the appearance of completeness and participants' attention often turns to minor details. At all points, participants should be encouraged and given avenues to critically reflect on the implication of the research results for their own work.

CONCLUSION

Although participatory design is often portrayed as a research orientation or a field, understanding it as a methodology leads us to better understand its promises and constraints, its limitations and its criteria—and, I think, also leads us to greater respect for the careful work that goes into developing a participatory design study. That's especially important for technical communicators. We are, after all, in a design-oriented field (Kaufer and Butler 1993) and we have drawn heavily on design-oriented research methodologies, methods, and techniques such as usability testing.

If we understand participatory design as an *orientation*, we are tempted to articulate a few general principles and retrofit our existing techniques to accommodate them. But if we understand it as a *methodology*, we are able to draw on a coherent body of methods and techniques operating within a general research design under common methodological premises. That is, we are able to conduct studies that have a great deal in common with

other studies; we are able to draw from and contribute to a coherent, common body of knowledge. Our work becomes relevant to others working in human-computer interaction, computer-supported cooperative work, and similar fields. **TC**

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Emerging Spaces in Community-based Participatory Design: Reflections from Two Case Studies

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ABSTRACT

This paper engages with issues of universality and locality in the context of community-based participatory design (PD), and focuses on the challenges and opportunities associated with incorporating local views and forms of participation in the design process. The notion of ‘designing for participation’ is advanced as a quintessential perspective for approaches in which design practices are re-configured from a community-centric standpoint. Building on insights from PD and community development studies, as well as on empirical evidence from two community design studies, we argue that designing for participation appears to be located in a space between the designer’s and local views of participation, which are at times both ambiguous and conflicting. To overcome these tensions, we argue for the importance of engaging critically and reflectively with PD in community contexts, and in this process capitalising on disciplinary dialogues that can expand the viewpoint from which PD projects are negotiated and evaluated.

Author Keywords

Community-based design, third space participatory design, design for participation, interdisciplinarity

ACM Classification Keywords

D.2.10 Design, Methodologies, Human Factors.

INTRODUCTION

More than a decade ago, Muller described participatory design (PD) as the third space in human-computer interaction, located between the “world of technology developers/researchers, and the world of the end-users” (2002:15). The underlying idea, further embraced by other PD theorists, is that PD may be conceived as a bridge between two worlds and two ways of knowing (Muller, 2002). Some of the significant conceptual and

methodological developments in PD build on this notion of bridging, whether taken as a space of encounter, of meeting halfway, or of exchanging knowledge and expertise.

With the entry of PD in new territories, such as local communities and developing contexts, new opportunities and challenges arise in enacting third space participatory design. Community-based PD appears to be one of the most challenging in this respect, due to knowledge gaps between designers and community members, binding local socio-cultural protocols (Sabiescu and Memarovic, 2013), and tensions between the designers’ and local epistemological frames (Bidwell and Hardy, 2009). Recent literature on PD in community contexts suggests that incorporating local understandings and patterns of participation in the design process is paramount for the success of such initiatives (Rodil et al., 2012; Winschiers-Teophilus et al., 2010; Bidwell and Hardy, 2009). On these grounds, community-centric PD questions the universality of the original Scandinavian PD approach and its array of associated design tools, methods and techniques (Elovaara et al., 2006; Puri et al., 2004). This opens the perspective towards “emergent and situated” (Verran and Christie, 2007: 215) design practices, in which the local community becomes the central reference point (Van der Velden, 2010).

This paper engages with issues of universality and locality in the context of third space participatory design in community settings, and focuses on the challenges and opportunities associated with incorporating local views and forms of participation in the design process. The notion of ‘designing for participation’ is advanced as a quintessential perspective for approaches in which design practices and methods are re-configured from a community-centric perspective. To examine the limits and challenges associated with designing for participation in community-based PD, the paper introduces two community design studies in different socio-cultural settings, located in Mozambique and Romania. Building on insights from PD, community-based design and community development studies, as well as empirical evidence from the two cases, we argue that designing for participation cannot be reduced to mere local views of participation. Designers carry with them their own views, whether implicit or explicit, and these have a fundamental bearing on how PD projects progress. Designing for

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participation appears to be located, rather, in a space between the designer's and local views of participation, which are at times both ambiguous and conflicting. Its translation in participatory design techniques, tools and processes fit for a local context is a challenging task, which needs to account for differences and tensions rooted in different ontological and epistemological frames.

To overcome these tensions, we elaborate on the concept of 'connectedness' from two standpoints. Firstly, we discuss the benefits of advancing community-based PD as a reflective endeavour on part of the designer, in full recognition of how PD both cultivates and is influenced by the interconnectedness, or intersubjectivity, of people. Thus, participation is always individual and always collective, and is always embedded in a broader 'web of significance' (Geertz, 1973). Secondly, we argue for the need to cultivate inter- and transdisciplinary dialogues beyond conventional boundaries, to help address complex societal problems. Community-based PD is approached as an illustrative example where disciplinary dialogues can enrich PD experiences by incorporating insights from community development studies. Ultimately, the goal of engaging in these dialogues is not to limit PD fluidity, but rather to stimulate critical reflection.

BACKGROUND

Participatory design with communities, in particular communities in rural and developing contexts, questions the universality of the original Scandinavian PD approach (Puri et al., 2004) and established notions about the role and nature of participation in design (Winschiers-Teophilus et al., 2010; Braa, 1996). This section presents an overview of PD studies in community contexts, singling out current debates around PD universality and locality, and changing notions of participation.

Community-centric Participatory Design

It has been argued that PD with communities, in particular indigenous or minority groups, does not rely on the application of the right method. Rather, it implies defining and negotiating design tools, methods and processes around the needs of a specific locality, including a contextual definition of local forms of participation (Winschiers-Teophilus et al., 2010). Successful collaborative design with communities entails design in the here and now, with designers and communities co-creating their own relevant tools and techniques, and agreeing on those processes that provide adequate responses and solutions to the situation at hand.

A notable feature of community-centric PD is the re-configuration of the design space, tools, and workflows by taking the local as central point of reference (Sabiescu, 2013). Maja van der Velden (2010) describes it as "design for the contact zone". The concept is borrowed from the work of Mary Louise Pratt (1991), who defines it as a social space of encounter between different cultural groups, characterised by exchange, but also clashes and conflict. Contact zones currently and historically characterise those contexts marked by colonialism and un-balanced power relations, which may produce unique practices and representations. For Van der Velden

(2010:12), design for the contact zone has no pre-defined articulations and rigid determinants, such as the fixed roles of 'designer' and 'user'. It is a localized practice in which joint actions and interactions contribute to shaping site-specific roles, tools, workflows and design concepts in a dynamic and evolving way. For example, Rodil and colleagues (2012:89-90) describe the process of co-designing a 3D visualisation space for indigenous knowledge in Namibia: *"...it was all a big melting pot: Where designer, the artefact and the space melt together within the design process, where the object of design becomes the tool for participation and where the designer becomes a part of the community and the community becomes designers."*

This process of local re-configuration can forge novel or re-interpret existing tools and techniques. For instance, Verran and Christie (2007:226) argue that video, a representational technology in the Western tradition, can be re-configured in indigenous contexts as performative act. A video-recorded testimonial can be enacted as a traditional Aboriginal performance, in which the storyteller draws live connections to tribe histories and geographies. This enables video as "technology of representation" to be appropriated and re-configured as "video as technologies of witness" (p. 221). This re-configuration marks a transition from a Western view of knowledge as representation, to the indigenous view embraced by the Yolngu aboriginal group, where knowledge is a re-enactment, a performative act circumscribed to a specific place and time (p. 219).

Local Notions of Participation

The different social and cultural values that communities espouse come to shape different views of participation that cannot be neglected in community-based PD projects. For instance, Puri and colleagues (2004) argue that there are no universal models of participation, giving examples of different if not opposed patterns of participation in South Africa and India. In a context such as the Xhosa-speaking rural communities of South Africa, decisions over activities that affect the entire community require consensus. This differs from certain rural contexts in India, where hierarchical models of communication and decision-making are predominant. On these grounds, it has been argued that forms and practices of participation in design should not be pre-defined, but rather configured dynamically through interaction with local communities, and shaped by local understandings, customs and practices (Winschiers-Teophilus et al., 2010). In some initiatives, interaction and participation patterns are modelled around local rites and practices. In their project with the Yolngu Aboriginals in Australia to build an indigenous knowledge system, Verran et al. (2006) have modelled activities on the local Garma ceremony, and positioned their approach to collaborative research and design as "performative knowledge-making".

Local Participation in a Community Development Optic

When working with communities, it is critical to question not only the forms of participation, but also how participation serves long-term community goals, such as

development and empowerment (Braa, 1996). There is by now an established body of work in development and community development studies that can provide insights in these matters. The concept of ‘development’ itself has evolved, from a Western-based vision focusing on economic growth and colonisation (cf. Escobar, 1992), to a focus on agency and proactive engagement of beneficiary groups in pursuing their own, self-designed development goals. This latter position is well represented by Amartya Sen’s definition of development as the “process of expanding the real freedoms that people enjoy”, including agency (or process freedom) and capabilities (or opportunity freedoms) (Sen, 1999). This approach places people rather than material progress at the centre of development efforts. Concepts such as ‘self-determination’, ‘voice’, and ‘empowerment’ have acquired renewed meanings in a developmental optic, and all call for an active role of beneficiary groups as the main agents and crafters of the better future they envisage for themselves. In this context, participation emerged in developmental actions as a viable way to design initiatives responsive to local needs, and allow the poor and disadvantaged to partake in decision-making on matters that affect them (Kanji and Greenwood, 2001).

Participatory approaches for development fall broadly in two categories: 1) the instrumental approach, where people’s engagement is seen as a means for gaining local support for a programme, to ensure acceptance and adoption, and hence to make it more effective; and 2) the transformative approach, where participation is associated with building capacity, local ownership and empowerment (Cleaver, 1999). These two approaches are associated with diverse tools and techniques (Kanji and Greenwood, 2001), and measure the performance of initiatives on different grounds. While the instrumental approach appears to be closer to the role of participation in traditional PD, the transformative approach compels us to broaden the perspective, and to question the reasons for involving communities in PD projects, and how they may benefit from this in the long term.

This paper focuses on those cases when PD is adopted in a community development perspective, and aims to shed light on how this perspective affects the creation of participatory spaces and the location of the designer and community members within. For this purpose, two PD studies in different socio-cultural settings, located in Mozambique and Romania, are introduced.

CASE DESCRIPTIONS

Romani Voices: PD for Cultural Representation

Romani Voices was a participatory project that aimed to explore the potential of digital technologies in giving voice to minority groups. ‘Voice’ was treated as a multidimensional concept that embodied ideals of empowerment and expansion of agency for pursuing self-designed goals for knowledge production and communication (drawing on Sen, 1999; Tacchi, 2008). The project involved the members of two Roma minority communities in rural Romania, who co-designed and co-developed digital communication artefacts that reflected their collective goals for knowledge production, representation, and communication. Hereafter, we focus

on one of these cases, involving the Kalderash Roma community in the village of Munteni, South-Eastern Romania.

Context

The Roma in Munteni are part of the Kalderash (or Coppersmiths) group, whose denomination comes from their traditional profession: the production of metal objects, such as spirits-brewing cauldrons, buckets, and pans. The Kalderash in Munteni used to be nomadic until the end of the 1950s, when Communist authorities forcefully settled them (Achim, 2004). Most Roma in Munteni continue to travel from spring to fall, moving further north and west searching for customers for their metal products, even if they have built homes in the village. Metal work is not only a traditional profession, but also the chief source of income for the Roma. Virtually all families in Munteni have at least one coppersmith among their members, whose income is used to support the entire family. It is a profession inherited and transferred for centuries, still alive, despite the professional restructuration caused by modernisation (Sabiescu, 2013). Other sources of income apart from metal work are scrap metal deals, roof painting services, soap making, and other small businesses. The uncertain and fluctuating revenue-making activities reinforce a life pattern based on immediate satisfaction of basic needs (Idem). With respect to its socio-cultural profile, the Roma community can be described as a pure type of *Gemeinschaft/Community* (Tönnies, 2002): a society characterised by a strong ‘collective self’, where collective goals have primacy over individual ones, and traditional customs and values are binding and unquestioned. Choices in an individual’s life are done to serve one’s family and one’s community. The Roma child is raised in respect of the tradition, and compelled to stand by it from a very early age. The family takes important decisions for their junior members, for instance the choice of profession or of bride/groom. The customs by which all members abide have been transferred and enacted for centuries, and are as strong at present as they were historically. Customs also regulate conduct and appearance according to age and social status. Orality is the predominant means of communication, and many members, especially women, are illiterate or semi-literate (Sabiescu, 2013).

The Roma have traditional leaders referred to as *bulibaşa*. At the time the project was conducted (2010-2012), the *bulibaşa* inherited this role from his father, the elderly *bulibaşa*. In recent years, when the local expert for the Roma was introduced in Romanian administration, the *bulibaşa* in Munteni assumed this role as well. In this role, the *bulibaşa*-mediator represents the Roma interests in the City Hall, and intercedes communication with the Roma (Sabiescu, 2013).

Methodology

The project employed a methodological framework based on ethnography and PD. The study lasted for 23 months, and was designed around intensive 1- or 2-week fieldwork and design activities, paced so that a gradual awareness of technology potentials and consequences could be developed in the local community. The first part

of the project was exploratory, in which several activities were devised to create a knowledge pool on three dimensions: 1) the local context, 2) available communication technology options, and 3) possible communication solutions relevant for the community goals. The local context was explored through ethnographic techniques, including interviews, focus groups and observations. Available technological options have been presented to local people through informative sessions and demonstrations. Communication solutions applied to the local context have been explored through content production demos, and the gathering and examination of cultural probes (Sabiescu, 2013). Several intermediary communication solutions (e.g. a community video blog) have been designed and produced to allow people to understand the potential of digital communication, and the consequences of using such solutions, especially for public outreach products. In this phase, it was agreed to build a community website. It took several months for people to develop a collective vision of their own website, which was lastly aimed to *“mirror people’s traditional lifestyle, difficult life conditions, and showcase their traditional metal work professions, through authentic testimonials of its members.”* (Sabiescu, 2013). The website was to be built through a participatory design approach involving local people, and populated with content produced by members.

Following a family-based content production approach, seven extended local families were subsequently involved in the participatory production of audio-visual content. The local leader recommended six of these families, while one joined later at the behest of the family chief. People used video and photo cameras to produce and gather stories and testimonials on themes ranging from cultural traditions to values, collective concerns and aspirations. Collective screening sessions were organised periodically in which recordings were visualised and commented upon. In this process, the most important community themes were tracked and validated based on collective consensus. Recordings were selected and edited to produce short movies for website publication.

In the last phase, a community website was designed and developed. Open card sorting was used to open the website design sessions. Participants grouped content samples and community themes to define the information architecture. In two subsequent design sessions, participants selected the content to be published, decided on content formats (e.g. stories, testimonials) and mapped it on the information architecture. The main outcome of the project was the community website (www.romanivoices.com). Due to low digital literacy, local people did not manifest an interest to manage the website, and an agreement has been set to keep the website online for three years after project completion.

RE-ACT: PD for Touristic Communication

This section describes a PD initiative carried out within the project RE-ACT (Social Representations and Actions for Improvement). RE-ACT was a collaboration project run by the Università della Svizzera italiana (USI Lugano, Switzerland) and the Universidade Eduardo Mondlane

(UEM, Mozambique), aiming to investigate how Community Multimedia Centres (CMCs) in Mozambique are perceived by different stakeholders, and consequently to devise activities for improving the range of services offered by CMCs to local communities (Rega et al., 2011). Overall, the RE-ACT project oversaw the development of nine initiatives to improve CMC services, involving CMCs and local communities in different Mozambican provinces (David et al., 2013). The co-design initiative described herein was carried out in the Island of Mozambique, and involved the affiliated CMC and members of the local community.

Context

The Island of Mozambique is located in the northeast of Mozambique in the Nampula province, which at present has 42 407 inhabitants, 14 889 out of them living on the island (INE, 2007). The island is a well-known touristic destination, and has a rich history, as it was the former capital of Mozambique. A number of NGOs and associations are based on the island, some of which offer basic computer courses to local communities. The CMC involved in the RE-ACT study was founded in 2007, and offers photocopying, typing and scanner facilities, as well as access to multimedia services. While it was initially intended to offer broader computer access, there was no working computer available at the beginning of the RE-ACT project (2011), and typing and email services were done from the laptop of the coordinator (Rega et al., 2011). The CMC also oversees a Community Radio, which broadcasts content in Portuguese and Emakhuwa, the local language (Ibid.). CMC users are mainly local inhabitants and students who need photocopies and email or typing services. At the time of conducting the fieldwork (2011), there were 16 volunteers working for the CMC.

In 1996, UNESCO declared the Island of Mozambique a World Heritage Site (WHS) due to its outstanding cultural value. As ratified by the World Heritage Convention, this called for increased attention to the WHS conservation, protection, as well as to its presentation, communication and creation of public awareness within the local community and the world at large (UNESCO, 1972). In this context, using the CMC resources and the RE-ACT project support, a set of actions were taken to conserve, protect and promote cultural and tourist activities in the Island of Mozambique. The forthcoming sections examine the steps taken for involving local stakeholders in the process of co-designing an ICT solution to promote tourism and cultural activities.

Methodology

The activities devised within the RE-ACT project had as general purpose to design and develop ICT solutions with inherent value for the communities where the CMCs were established (David et al., 2013). In the Island of Mozambique, the RE-ACT project provided an opportunity to improve relations between the CMC and the local community, hence contributing to the provision of novel and relevant CMC services. Two main goals were settled: 1) Co-design an ICT solution to communicate the social, cultural, and touristic value of

the Island of Mozambique to national and international tourists and travel agents; and 2) Enable the local CMC to provide a novel range of services, which would create a link between the CMC and the community.

The CMC was entrusted with leading the design activities, overseeing the implementation of the ICT solution, and managing relations with the other local and international stakeholders. The CMC selected key figures to represent each social group of the local community (i.e. housewives, teachers, local business entrepreneurs, local government entities) and mediate relations between the community and the RE-ACT team.

The design of the ICT solution was conducted by means of two co-design workshops. The first workshop was devised to create initial concepts for ICT solutions for each of the nine CMCs and affiliated communities involved in the RE-ACT project. The workshop sessions were programmed to support a co-design process, and included spaces for information sharing, brainstorming, debate, and decision-making. The workshop had 32 participants, including the CMC directors, the RE-ACT team, and a network of people related to the establishment and administration of CMCs in Mozambique. Participants shared ideas and drafted proposals for ICT-supported products and services that could benefit local communities, and capitalise on the social and technical infrastructure of each CMC. In this workshop, it was decided that the Island of Mozambique would benefit from the creation of a website for promoting the social and cultural value of the island to touristic audiences.

The second co-design workshop was held on the island itself. It lasted three days, and had 12 participants, including the CMC staff, representatives of the community (housewives, local entrepreneurs connected to tourism activities, the representative of the island association for tourism and the representative of the fishermen association), local government entities, and the RE-ACT team. This workshop had as goal to present to the local community the improvement action conceived in the first workshop, negotiate its relevance and feasibility, and on this basis proceed to the design of the ICT solution. The community representatives had the power to reject or accept the concept, and further shape it to better fit their needs.

The workshop was divided in three parts. During the first part, the CMC director presented the idea of building a website for promoting the island as a WHS. The local stakeholders brought changes to this concept, arguing that the website should promote only local tourist actors, and exclude international touristic operators. During the second part of the workshop, community representatives suggested that content gathering should be managed by the community and later revised by the RE-ACT team. The RE-ACT team was also asked to mitigate the gap in technology skills by providing staff training to the CMC (online and face-to-face). Initial training in website management was provided to eight of the CMC staff members during the workshop. The third part was dedicated to gathering requirements, drafting a

provisional information architecture for the website, and planning the collection of materials for content production. Roles and responsibilities were distributed among stakeholders. The CMC staff was the leading figure, responsible with overseeing the design and implementation of the website. Representatives of the community were to remain in constant contact with the CMC staff and be consulted at key stages during the website implementation. The RE-ACT team was in charge with the website technical implementation and providing technical consultancy.

The website (www.ilhademocambique.org) was received well by the local community and the CMC. The CMC was entrusted with the responsibility to maintain and update the website, while the RE-ACT team remained available for consultancy on technical issues.

DATA ANALYSIS

To understand how the notions of participation embraced in each project shaped the PD approaches adopted, the data available from each case was assessed against reflections from designers involved in each community project. In Romani Voices, data was available from multiple sources including the designer's observations, interviews, focus groups, and cultural probes gathered at different moments during the project course. In the RE-ACT case, the analysis was done on a data corpus drawn from observations during the design sessions, three-monthly reports submitted by local stakeholders, and interviews with CMC staff and local community members after the website was launched.

Designers involved in each case study were asked to reflect on how they viewed participation before entering the field, and how their views evolved as the project progressed. They identified landmark events and issues that challenged or changed their views, as well as challenging moments in the project, which they associated with different understandings of participation.

DESIGNING FOR PARTICIPATION

The two projects were characterised by genuine commitments to incorporate local forms of participation, and infuse them in the design course. At the same time, designers did not approach communities with no prior expectations. They had embarked in the initiatives with their own views on the role of participation for meeting design objectives. This section highlights how hybrid visions of participation, lying at the convergence between an external and a local point of view, were gradually instantiated in context-specific design approaches. Despite the efforts to "go local", these spaces were not deprived of antagonism and cultural conflict, and challenged the grounds for a participatory approach.

Designers' Visions of Participation

From the accounts shared by the designers, local participation was expected to contribute towards either project effectiveness or the empowerment of local stakeholders. In the case of the former, participation was seen as an instrument for designing an artefact that met fully the project goals and the views of the local stakeholders. In the case of the latter, participation was thought to enhance varied attributes or qualities of local

stakeholders, such as knowledge, skills, self-confidence, technical abilities, etc.

A pronounced orientation towards participation as a means for ensuring project effectiveness was evident in the Romani Voices study. Local participation was meant to contribute to building a culturally representative artefact, referring to the quality of an artefact to reflect local ways of knowing and being (Manovich, 2001). At the same time, it was acknowledged that effectiveness was closely connected with empowerment. To participate effectively in the project, people needed to enhance their levels of knowledge, understanding, proactivity, and confidence in providing inputs. At the same time, a key project goal was to cultivate capacity for voice and expression that could potentially manifest in the participants' future beyond project completion. The Romani Voices designer therefore employed a dual optic, wherefrom participation for effectiveness could not be separated from participation for empowerment, and both depended on developing people's skills, awareness and confidence.

In the RE-ACT case, participation was seen as a means to empower the local CMC to continue to manage the website and propose other ICT-supported activities for the local community after the project had ended. Building the website was not only a one-time design and development exercise, but also a venture in which the CMC needed to garner the necessary expertise and nurture the required community relations towards creating a long-lasting link.

Local Visions of Participation

The two localities present different, almost opposed ideas of participation: hierarchical models and bounding rules of conduct in the Kalderash Roma community, and democratic ideals of participation in the Island of Mozambique.

The Kalderash Roma, as many other traditional Romani groups, present patriarchal and hierarchic forms of social organisation. Strict rules of conduct define intra-community interaction and communication, with well-defined patterns and taboos based on gender, age and social status. For example, the young do not speak against the elderly in group contexts, and the same goes for women when men are present. These rules also guide communications with people from the outside, especially in cases where interactions have a formal nature, such as the Romani Voices project. The community leader, the *bulibaşa*, acts as mediator and representative of community interests in interactions with outsiders. Some of these rules and patterns were obvious from the first community meetings, while others emerged gradually as the project progressed, fundamentally shaping local participation in the project.

The Mozambican case depicts very different visions of participation, characterised by ideals of equality and shared decision-making. These views are rooted in democratic and participatory values espoused by the culture of the island. They are embraced by island inhabitants, and permeate relations between members as well as relations with 'outsiders'. In working as a

mediator for the community, the CMC confirmed its commitment to these values, and was expected to act transparently and be open to community inputs.

Instantiation in Participatory Design Approaches

The perspectives on participation embraced by designers and local people converged in the creation of distinct patterns of local participation in each case study.

In Romani Voices, designers' beliefs about the purpose of participation influenced the project macro-structure, tools and techniques, and landmarks in the way the project progressed. This was underpinned in the belief that, in order for participation to be effective, people had to be empowered to contribute, decide, and act in confidence. To meet this goal, the project dedicated significant time to build knowledge and skills before embarking on the actual website design, which was done only in the 16th month of the project. The activities before that time were devised to support knowledge building and familiarise people with the language of digital media and the Internet. The tools and techniques used were adapted to members' low digital literacy, and favoured hands-on activities, demonstrations, and critical analysis of the digital media artefacts produced.

Local understandings of participation shaped, by contrast, the format of activities, micro-moments in the project course, as well as inclusion and exclusion scenarios. *First*, they enacted a pattern for decision-making that reflected a hierarchical model. The community leader and representative, the *bulibaşa*, was the most important figure in this respect. In his position of community representative and mediator, he was invested with the power to take decisions over matters of collective concern. *Second*, local customs regulated who could intervene, speak, and debate in collective sessions. For example, when the *bulibaşa* was present, people would refrain from uttering views that contradicted his position, or would tend to comply. When men were present, women generally kept a low profile, and could become much more proactive when men or elderly people were not around. It should be underlined, however, that these taboos were not impediments to engaging different genders and age groups, but rather required sensitivity to interacting with local people in different group compositions. At times, women may shy away from intervention or training. For example one young woman refused to touch the video camera motivating that she was not able to do this. At the same time, one other local woman became one of the most important, imaginative and hard-working content producers of her own initiative, after learning to use the camera initially entrusted to her husband. *Third*, local patterns of relatedness and alliances fashioned inclusion and exclusion scenarios, and defined who could be involved in the project and how. The leader generally mediated project meetings and introduced the designer to social circles and extended family clans. Contact with most of these circles continued for the project duration.

In the RE-ACT case, designers viewed participation as a means for empowering the CMC and enhancing its capacity to act as representative of the local community

through online media in the long term. The participatory strategy was structured along two elements. First, the CMC was entrusted with the role of leader, manager and mediator in all issues that involved the local community. In this process, the RE-ACT team assumed the role of advisor and consultant. Second, special activities were devised to transfer expertise to the CMC for website management. This was done by incorporating training sessions during the second design workshop, and organising further training sessions during the more advanced stages of website development.

Local ideas of participation, valuing democratic decision-making, marked the format of the design sessions. These were characterised by open sharing of information and equal opportunities to give input and make decisions. For instance, during the second co-design workshop, local community representatives had the power to accept, refuse or modify the idea of creating a promotional website. Based on their inputs, the website concept was modified to promote only local businesses, excluding international touristic chains operating on the island.

Challenges in the Local Appropriation of the Design Space

Despite the openness to embrace local notions of participation, both projects were marked by challenges and contestations, some of which indicate tensions between local and external views of participation.

In Romani Voices, the designer deemed people's participation essential for building a website that they would consider their own. Hierarchical models of decision-making, communication and relatedness at times obstructed this. For instance, people were shy interveners and refrained from putting forward determinate views, driven by lack of confidence in the importance of their inputs, and beliefs that the leader or people higher in social status should be involved. The local leader could decide and intervene in matters where the designer thought broader participation was needed. This aspect was most evident when discussing ethical issues. Since its inception, the project embraced notions of ethics based on full prior informed consent, and devised tools for meeting ethical measures at collective and individual level. Collective research agreements as well as individual consent forms were drafted. However, the leader specified that by signing the collective agreement he implicitly decided on behalf of all the members, therefore there was no need for delivering individual consent forms. Implicit rules also determined who could communicate and give input in specific situations. It took time for the designer to understand that interventions from people were conditioned by who was present in a group meeting.

There were instances, in Romani Voices, when it became apparent that clashes were not only due to ill familiarity with local customs, but were underpinned by fundamentally different visions and epistemologies. One incident betraying such difference occurred during the first website design session. During this session, participants were asked to group content samples (printed on paper and playable as video) to create website

categories and tags. After initial difficulty, the designer made some examples, grouping content according to what he considered logical; for instance grouping two local customs videos in the category 'Traditions'. After two such examples, the local leader grouped five pieces of content, dealing respectively with child education problems, life in the tent, lack of work places, living on the road, and poverty and everyday life. They were all connected, he mentioned, as they were all outcomes and effects of poverty. His vision, rooted in a core community concern, was embraced by the others and was adopted when the final information architecture was defined.

In RE-ACT, the designers' ideal of participation for empowerment, in which leadership was vested with the CMC, was against democratic notions of inclusion as espoused by the local community. The leading role of the CMC was challenged on several grounds. The number of business entrepreneurs involved, initially 28, was reduced to six. While there were many factors determining this decrease, certain entrepreneurs indicated that their exclusion was a political act. Certain people also felt that the project would have been run more effectively if RE-ACT had managed it, leaving a marginal, mediatory role for the CMC staff. The CMC staff was not initially confident in their ability to maintain the website. Early during the second workshop, one CMC member commented: *"Although the CMC staff director and the community want to implement a website we are not sure that we can manage such a platform. Because we don't have training in website management, only in radio and teaching computers."* Despite training provision, the lack of long-term familiarity with website management platforms undermined the confidence of CMC staff in their ability to oversee the technical aspects of maintenance without the support of the RE-ACT team.

DISCUSSION AND CONCLUSIONS

This paper examines the foundations that underpin 'designing for participation' in community contexts. The analysis of both studies brings forth a series of aspects around the promises and limits of this perspective, and points to further directions in developing such an approach. These are discussed below along four lines of argument, namely (i) enacting local notions of participation, (ii) contested spaces of encounter and the reflective designer, (iii) autonomous local actors and power differentials, and (iv) inter- and transdisciplinary dialogues.

Enacting Local Notions of Participation

The analysis of the two cases indicates that the views on participation held by designers and local communities were not only different in substance, but were also focused on different aspects. Designers were primarily concerned with the purpose or role that participation served in the frame of the project or for longer-term community development. Local customs and values, on the other hand, dictated the forms that participation could take: who could be involved and how, how were decisions taken, how was collective consensus achieved. The two perspectives appeared to shape different facets in the creation of participatory spaces. Considerations on the purpose of participation determined the macro-structure

of the projects, and were more influential in dictating the flow and activities. Local customs, on the other hand, shaped the format of participatory sessions, micro-moments in these sessions, and inclusion and exclusion scenarios. While the two studies are not necessarily emblematic for the variety of approaches in community-based PD, we argue that notions of localisation in community-based PD studies should include precise considerations of both the limits and the articulations of the spaces, tools, and processes that are contextualised. In this respect, some important questions can be asked: *At what level does/should PD incorporate local views? Can we speak of incorporating local views when local forms of participation are integrated in a design process framed by an external logic? Or should the entire design apparatus be deconstructed and reconstructed around an indigenous logic and a local reference point?* These questions reflect debates in anthropology and community-based participatory research, around the tensions between emic (inward) and etic (outsider) perspectives (Guba and Lincoln, 1994). If project activities emulate local customs and ways of being at superficial levels, it is likely that disparities and distortive interpretations will thrive. Bidwell and Hardy (2009) have argued that even when participation is rooted in local ways of expression (e.g. through drama, performance, storytelling), important articulations of the design process, such as the elicitation of requirements, are still being done from a Western-centric perspective, which acts as a distortive interpretive lens.

Working in an indigenous logic is, on the other hand, a challenging task. Theories that support this approach in design have been adopted from several social sciences disciplines. For instance, the notion of placing “an ethnic group at the centre of the inquiry” (Tillman, 2002: 4), or Smith’s concept of “reconstructed sociology”, which positions the sociological inquiry from the standpoint of a group or consciousness situated in a particular time and space (Smith, 1999: 74). Yet, few empirical studies to date have stories, strategies, and ideas to share from the field. Some of these suggest that long-term immersion in the field, enduring partnerships with communities, and effective reciprocal knowledge exchange are essential elements for this purpose (Bidwell et al., 2013; Winschiers-Teophilus et al., 2010). These strategies give the community time and opportunities to develop design thinking by long-term and cyclical design exercises that probe and analyse design ideas coming from both sides (Winschiers-Teophilus et al., 2010, 2013). This long-term engagement is also a means to overcome power differentials and increase the confidence of local community members to engage proactively in the creation or critical assessment of design spaces.

Contested spaces of encounter and the reflective designer

The two cases presented suggest that, despite efforts to embrace local notions of participation, the actual design of participatory spaces is the product of an encounter. The articulation of participatory spaces is therefore often fragmented and limited, in its emergence as a unique outcome of dual visions. While these visions could

ideally converge in a meta-space of encounter and dialogue, the two cases and others in the community-based PD literature point to important challenges arising from their design and implementation. Some issues are contingent on the premises of externally initiated projects. Having been proposed from outside the community, these projects raise issues and open questions with respect to local benefits, building local ownership, cultivating agency and considering long-term community development goals. For instance, whose empowerment goals are actually considered in projects that seek specifically to cultivate empowerment, as was the case for the two studies described? Whenever the balance of power inclines towards the external team, paternalistic attitudes can be developed despite good intentions, leading to a reinforcement of patterns of dependency (Freire, 2006). The Romani Voices study is an interesting example of tensions between external visions of empowerment and local customs that valued hierarchy and maintained strict taboos for social interaction and decision-making. Ultimately, the case raises the question as to whether empowerment goals can be actuated in hierarchical settings without affecting local dynamics and traditional ethos. The case does not provide a solution to dealing with such issues, but rather points to the need for increased sensitivity and self-reflection on part of the designer, so that even ideal standards are not taken for granted and proposed as unique solutions.

Other factors of contention in the articulation of participatory spaces reside in deep differences between designers and local people, ranging from epistemologies, meaning and value attribution, to gaps in knowledge and skills, especially technological skills. The meaning of technology itself is likely to be framed from very different visions, fashioned by peculiar patterns of interaction with different kinds of technology. Specific activities can be devised to bridge some of these gaps, for instance hands-on training, informative sessions and demonstrations. Yet, others are deep-seated and may continue to affect the pace and flow of the design continuum, leading to tensions, fragmented understandings and missed steps whose actual roots are not immediately obvious. Time (Winschiers-Teophilus et al., 2010) and dialogue (Sabiescu, 2013) are some of the aspects mentioned in the literature for dealing with such issues. We argue, moreover for the need to cultivate an awareness of intersubjectivity and interconnectedness as the larger canvas for conceiving participation patterns. The community’s shared culture, languages, past traditions and contemporary expressions all affect understandings and forms of participation. At the same time, designers are not neutral figures and bring their own understandings and experiential luggage, which are seldom explicitly dealt with. We therefore argue that for a thorough understanding of the underpinnings of participation, and for skilfully deploying means for designing participatory spaces, there is a need to engage in community-based PD as reflective practitioners (Schön, 1991). The premises for the reflective practitioner/designer have been sketched in some of the most influential writings about the participatory inquiry

paradigm. As argued by Heron and Reason (1997), any conceptual construct is validated by consensus of specific communities and set in a wider experiential context. Critical subjectivity, attending to action with a “critical consciousness” and “self-reflexive attention” (Idem:282), is necessary for the designer to act while fully aware of the experiential roots and determinants of their own thinking. At the same time, critical subjectivity is to be cultivated in exchanges, shared spaces and dialogues, in which the designer is simultaneously a facilitator and convenor. Critical subjectivity thus extends to critical intersubjectivity, using shared experience, dialogue and feedback to engage in action with a critical consciousness of its underlying experiential determinants and limits (Idem:283).

Autonomous Local Actors and Power Differentials

The Mozambican case is a peculiar instance of PD, as full autonomy was vested with local actors, while the design team served as advisors over management and technical issues. The expectation was that the CMC would assume responsibility and build capacity in the long term. This approach was rooted in a developmental optic, by which local ownership of a project is thought to be positively related with empowerment (Cleaver, 1999) and project sustainability. Design initiatives in which local members are vested with leadership roles have the potential to contribute to community-driven design, where members grow and manage their own information systems and infrastructures without relying on external support (Karasti and Baker, 2008). Successful participatory design initiatives of this kind have been reported in the literature (e.g. Merkel et al., 2004).

Yet, in our study, despite well-intentioned activities for transferring expertise, and the availability of the design team throughout the project, the empowerment goals did not appear to be fully met. The analysis offered some surprising insights in this respect. Several community members were not pleased with how the process was managed, and preferred for the RE-ACT team to direct the project. This indicates that vesting project ownership in the hands of local stakeholders is not necessarily a sure way to success. And as our study indicates, the aspects that should be better managed are not related to transfer of skills and expertise, but rather to power, responsibility, and human relations. Despite good intentions, the RE-ACT project appeared to affect the balance of power between the community and the CMC, and introduce an external power differential that marked their relations. These insights confirm, as suggested by previous studies (e.g. Bidwell and Hardy, 2009), the importance of considering early in the advancement of a PD project how a design intervention is positioned within and may potentially affect local power structures.

Inter- and Transdisciplinary Dialogues

Community-based PD can be said to be fundamentally different from traditional workplace PD in at least two respects: it deals with culture and it deals with development. In community contexts, PD engages a group of people with a common history and an imaginable collective future. The PD project is bound to be affected by how local people think, act, communicate

and relate to each other. At the same time, the PD project will also affect some aspects of the local relations and power structures, marginally or fundamentally. In short, PD does not work in a vacuum, and needs to both consider and be accountable for its role along the cultural and developmental continuum within the community. We argue that, just as PD does not work in a vacuum in real community-based projects, there is a need for locating PD practices in inter- and transdisciplinary discourses about culture and development. In the present cases, designers adopted a community development approach and, as evident in their self-reflections, this had an observable effect on how they viewed participatory design practices. Our contention is that community-based PD can use insights, discourses, and even frames from other community-focused disciplines. This may help to address tensions in the early stages of PD, or to explore the implications of PD in its evaluation stages. While some studies in the PD literature actively engage with these disciplines (e.g. Braa, 1996; Puri, 2004), we propose that inter- and transdisciplinary dialogues could become more common and infuse design thinking throughout PD projects.

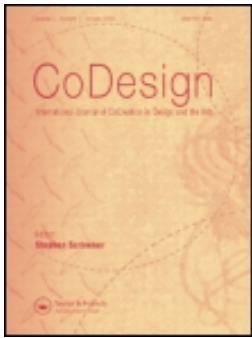
To contribute to the inter- and transdisciplinary undertakings of PD, and to help transcend its multimodal existence, we may envisage a series of “actionable strategies” as research agendas for (community-based) PD (adapted from Hassan, 2011:456): (1) agree on the intellectual ideals of PD, lending credence to emerging spaces; (2) conceptualise the primary directives (e.g. time, learning, power) in participatory endeavours; (3) construct and develop new theories (e.g. emic/etic and third space design; design for participation); (4) establish disciplinary properties and seek unified solutions to complex design issues; and thus (5) contribute to a discourse for creating new domains of PD, which may offer a unique disciplinary subject matter. It is hoped that these agendas contribute to the transdisciplinary emergence of participatory design, as such deeming it an increasingly influential field. These dialogues and an expanding knowledge base do not preclude advancing a PD initiative that is open, context-responsive, and flexible. They serve, rather, to articulate designers’ position in the field, in the process becoming facilitators for local people to render explicit their own position.

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Co-creation and the new landscapes of design

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Designers have been moving increasingly closer to the future users of what they design and the next new thing in the changing landscape of design research has become co-designing with your users. But co-designing is actually not new at all, having taken distinctly different paths in the US and in Europe.

The evolution in design research from a user-centred approach to co-designing is changing the roles of the designer, the researcher and the person formerly known as the ‘user’. The implications of this shift for the education of designers and researchers are enormous. The evolution in design research from a user-centred approach to co-designing is changing the landscape of design practice as well, creating new domains of collective creativity. It is hoped that this evolution will support a transformation toward more sustainable ways of living in the future.

Keywords: participatory design; design research; co-design; co-creation; collective creativity; user-centred design

1. Introduction

Over the past six decades, designers have been moving increasingly closer to the future users of what they design. Especially in areas where technologies mature, and the next new feature is no longer of value, manufacturing companies have been increasingly open to approaches that define the product based on what people need. The first advances, well consolidated now in industrial practice and education, practiced user-centred design from an ‘expert perspective’ in which trained researchers observe and/or interview largely passive users, whose contribution is to perform instructed tasks and/or to give their opinions about product concepts that were generated by others. The user-centred design approach (i.e. ‘user as subject’) has been primarily a US-driven phenomenon. Increasingly, since the 1970s, people have been given more influence and room for initiative in roles where they provide expertise and participate in the informing, ideating, and conceptualising activities in the early design phases. The participatory approach (i.e. ‘user as partner’) has been led by Northern Europeans. The two approaches are now beginning to influence one another. Figure 1 gives an overview of the current state of the human-centred design (research) landscape (discussed in more depth in Sanders 2006a).

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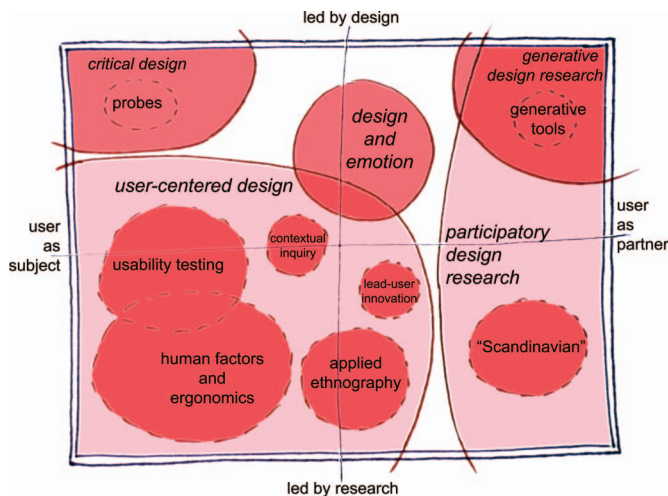


Figure 1. The current landscape of human-centered design research as practiced in the design and development of products and services.

Within this landscape, in the area of participatory design, the notions of co-creation and co-design have been growing. The terms *co-design* and *co-creation* are today often confused and/or treated synonymously with one another. Opinions about who should be involved in these collective acts of creativity, when, and in what role vary widely. Online dictionaries do not yet have entries for *co-creation*, *cocreation*, *codesign* or *co-design*. Wikipedia, the online encyclopaedia, has only preliminary entries on *co-creation* and *co-design*.

The authors take *co-creation* to refer to any act of collective creativity, i.e. creativity that is shared by two or more people. Co-creation is a very broad term with applications ranging from the physical to the metaphysical and from the material to the spiritual, as can be seen by the output of search engines. By *co-design* we indicate collective creativity as it is applied across the whole span of a design process, as was intended by the name of this journal. Thus, co-design is a specific instance of co-creation. Co-design refers, for some people, to the collective creativity of collaborating designers. We use co-design in a broader sense to refer to the creativity of designers and people not trained in design working together in the design development process.

Figure 2 shows a simple representation of the design process today. Of note is the large and growing emphasis on the front end. Formerly called ‘pre-design’, the front end describes the many activities that take place in order to inform and inspire the exploration

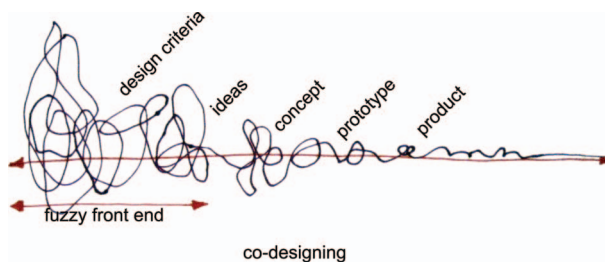


Figure 2. The front end of the design process has been growing as designers move closer to the future users of what they design.

of open-ended questions such as ‘how can we improve the quality of life for people living with a chronic illness?’, or ‘what is the next big thing in family leisure time?’ The front end is often referred to as ‘fuzzy’ because of the ambiguity and chaotic nature that characterise it. In the fuzzy front end, it is often not known whether the deliverable of the design process will be a product, a service, an interface, a building, etc. Considerations of many natures come together in this increasingly critical phase, e.g. understanding of users and contexts of use, exploration and selection of technological opportunities such as new materials and information technologies, etc. (Stappers 2006). The goal of the explorations in the front end is to determine what is to be designed and sometimes what should not be designed and manufactured. The fuzzy front end is followed by the traditional design process where the resulting ideas for product, service, interface, etc., are developed first into concepts, and then into prototypes that are refined on the basis of the feedback of future users.

2. A quick glance at history

It is interesting to note that even though the terms *co-creation*, *cocreation*, *codesign* and *co-design* have not yet made much of an impact on the online dictionaries and encyclopaedia, each of these terms retrieves hundreds of thousands of hits on the search engine www.google.com. In fact, *co-design* alone pulled up 1,700,000 hits on 2 July 2007. This attests to the interest of people around the world in these content areas. Moreover, counts on Google Scholar (11,800 for *codesign*, 538 for *co-creation*, as compared to 3,470,000 for *product design* and 17,400,000 for *design*; on 18 August 2007) indicate that the terms are also seriously discussed in academic design circles. It seems that we are talking about a very recent and relevant phenomenon. Or are we?

Actually, the practice of collective creativity in design has been around for nearly 40 years, going under the name *participatory design*. Much of the activity in participatory design (this was the terminology used until the recent obsession with what is now called co-creation/co-design) has been going on in Europe. Research projects on user participation in systems development date back to the 1970s. In Norway, Sweden and Denmark the Collective Resource Approach was established to increase the value of industrial production by engaging workers in the development of new systems for the workplace. The approach put together the expertise of the systems designers/researchers and the situated expertise of the people whose work was to be impacted by the change. The approach, thus, built on the workers’ own experiences and provided them with the resources to be able to act in their current situation (Bødker 1996).

The beginnings of the participatory design movement were occurring elsewhere at the same time. There is a wonderful, but unfortunately out-of-print, book edited by Nigel Cross and made up of papers that were presented at a conference called *Design Participation*, held by the Design Research Society in Manchester, England in September of 1971. The papers came from contributors who were at the time teaching and practicing in the fields of economics, design, architecture, planning, building science, design research, and mechanical engineering.

In the Preface to *Design Participation*, Cross (1972) wrote:

professional designers in every field have failed in their assumed responsibility to predict and to design-out the adverse effects of their projects. These harmful side effects can no longer be tolerated and regarded as inevitable if we are to survive the future . . . There is certainly a need for new approaches to design if we are to arrest the escalating problems of the man-made world and citizen participation in decision making could possibly provide a necessary reorientation. Hence this conference theme of ‘user participation in design’. (p. 11)

In the Closing Comments for *Design Participation*, Robert Jungk, a futurist and social inventor, advocated:

we could talk not (only) about participation at the moment of decision but about participation at the moment of idea generation . . . We can begin the preparation for this radical change. As a prognostician, I don't think this change will take place before the end of the century. We will have to suffer first from the lack of foresight of our fathers and forefathers. After that, something radically different can come, but it won't come on its own: it has to be prepared. (p. 122 in Cross 1972)

We can see now, well into the next century, that Jungk was correct in predicting that change would not come for many years. Hopefully we are well enough prepared for such change. It is interesting to note that participation in the design process, as it is practiced today, is focused more on the exploration and identification of presumably positive future opportunities than it is on the identification and amelioration of adverse consequences. However, both perspectives will need to be addressed in an integrated way if we are to face the immense challenges of living in the 21st century.

Co-design as it is practiced and discussed today takes on quite different manifestations, depending upon the expertise and mindsets of its practitioners. A few of the key proponents of co-design are briefly described below. It is of interest to note that the best known proponents of co-design originate from business or marketing and not from design practice.

C. K. Prahalad and Venkat Ramaswamy are usually given credit for bringing co-creation to the minds of those in the business community with the 2004 publication of their book, *The Future of Competition: Co-Creating Unique Value with Customers*. They propose:

The meaning of value and the process of value creation are rapidly shifting from a product- and firm-centric view to personalized consumer experiences. Informed, networked, empowered and active consumers are increasingly co-creating value with the firm. (Prahalad and Ramaswamy 2004)

Frank Piller also writes extensively about the co-creation of value between companies and customers. Piller is considered the expert on the concept of mass customisation, having analysed hundreds of companies pioneering this approach to selling their goods (Tseng and Piller 2003). Eric von Hippel (2005) works with 'lead users' in co-creative activities. Lead users are people who have already explored innovative ways to get things done and who are willing to share their approaches with others. Patricia Seybold (2006) also works with 'lead customers', the small percentage of customers who are 'truly' creative. The von Hippel and Seybold approaches to co-creation limit participation in the design process to an elite and very carefully selected group of people. It is not yet clear whether these elite groups of people can represent and speak for the majority of people who will actually use the goods and services that are being designed and developed.

Co-creation is by now being touted at all points along the product development process, particularly in the later stages. Websites such as www.NIKEiD.com allow people to customise their own shoes, for example, by choosing colors and detailing. For many, co-creation is the latest trend in marketing and brand development. It is the latest of ways to get new products and services into an already overcrowded marketplace. For example, it is being espoused as a powerful new tool for product naming, packaging, promoting and advertising.

In our experience as researchers and practitioners we have seen that co-creation practiced at the early front end of the design development process can have an impact with positive, long-range consequences. This mirrors Jungk's observation that 'participation at the moment of idea generation' is an important place to be practicing participatory design. However, 'participation at the moment of decision' is gaining in interest as well. The application of participatory design practices (both at the moment of idea generation and continuing throughout the design process at all key moments of decision) to very large-scale problems will change design and may change the world. Participatory design has the potential, as Cross described, 'to arrest the escalating problems of the man-made world'. In fact, the co-creative paradigm is being espoused by those exploring the intersection of science and spirituality as one of the primary means for transformation of the dominant worldview that is taking place today. This transformation is from an unsustainable way of life to one 'in which both science and spirituality reconfigure our most basic understandings of human consciousness and how to live harmoniously in a healthy and sustainable ecosphere.' (Institute of Noetic Sciences 2007)

2.1. *Why has it taken so long?*

There are a number of reasons why it has taken so long for the principles and practices of participatory design/co-designing to make an impact on the man-made world. First, to embrace co-creativity requires that one believes that all people are creative. This is not a commonly accepted belief, particularly amongst those in the business community. In fact, the primary business-driven approaches to co-creation (i.e. von Hippel 2005 and Seybold 2006) are based on the assumption that only 'lead' people can become co-designers. Embracing participatory thinking flies in the face of the 'expert' mindset that is so prevalent in business today. The existing power structures in companies are built on hierarchy and control. Co-designing threatens the existing power structures by requiring that control be relinquished and given to potential customers, consumers or end-users. It is very difficult for those who have been successful while being in control to give it up now or to imagine a new way of doing business that can also be successful. The new generations are having an easier time in distributing and sharing the control and ownership. This change in attitude is largely possible because the internet has given a voice to people who were previously not even a part of the conversations. But it will take some time before the predominant culture accepts egalitarian idea sharing. It is also difficult for many people to believe that they are creative and to behave accordingly. Co-designing requires creative initiative on the part of the entire team: researchers, designers, clients and the people who will ultimately benefit from the co-designing experience.

Secondly, participatory thinking is antithetical to consumerism, in which personal happiness is equated with purchasing and consuming material goods. Hopefully, consumerism and the rampant consumption that goes with it has almost run its course. In many parts of the world, the needs that capitalism has worked so hard to meet have been met and so new needs are now being invented. Meanwhile, in other parts of the world, basic human needs (e.g. clean water) are not met. In our studies over the last 10 years we encounter that, increasingly, people want a balance between passive consumption and the ability to actively choose what kinds of more creative experiences to engage in and how (see Sanders 2006b). Unfortunately, it will still take years for the culture to shift away from consumerism towards the consumptive/creative balance that people seek. The renewed interest in sustainable practices is also helping to fuel that fire.

A third reason that it has taken co-creation so long to have an impact is that participatory design has been seen as academic endeavour with little or no relevance for the competitive marketplace. In many parts of industry, investment in research is looked upon as a non-obvious step, investment in user studies a big and expensive step, and user participation a radical step into the unknown. This is beginning to change now as product development becomes increasingly knowledge-intensive, and industries and universities look to each other for collaborative explorations in innovation.

Last but not least, the relationships between new technologies and future human experiences have just recently become very complex and integrated. In the past, the manufacturing industries were characterised as being either manufacturing-driven or, more lately, technology-driven. The move of companies to pay attention to ‘user experience’ is in part motivated by stagnation of the technology push. For example, as car manufacturers find it hard to compete on technical quality and price, they are forced to look outside the product to the user and his/her context. They are becoming aware that the car driving experience is now of primary concern.

2.2. *What is going on in design practice?*

Design practice has been influenced by the changing landscape of human-centred design research. The user-centred design approach, which began in the 1970s and became widespread by the 1990s, proved to be most useful in the design and development of consumer products (Sanders 1992). But it is now becoming apparent that the user-centred design approach cannot address the scale or the complexity of the challenges we face today. We are no longer simply designing products for users. We are designing for the future experiences of people, communities and cultures who now are connected and informed in ways that were unimaginable even 10 years ago.

Consequently, new disciplines of design have begun to emerge. ‘Interaction design’ was first introduced in the late 1980s by Bill Moggridge and Bill Verplank (Moggridge 2007). Interaction design is just now beginning to be offered as a field of study in the curricula of design programs at universities. ‘Service design’ started to receive attention in 2006 with the advent of the first service design conference, *Emergence* 2006, that was put on by Carnegie Mellon University’s School of Design. ‘Transformation design’ was introduced in 2006 in a White Paper by that name published by the UK’s Design Council. Each of these new disciplines incorporates several of the traditional design disciplines within it. For example, service design integrates visual communication design, information design and interaction design. Transformation design, the newest of the emergent design disciplines, is based on participatory practices in combination with user-centred methods. It ‘builds on traditional design skills to address social and economic issues. It uses the design process as a means to enable a wide range of disciplines and stakeholders to collaborate’ (Burns *et al.* 2006).

A snapshot in time (see Table 1) shows that we are moving from the design of categories of ‘products’ to designing for people’s purposes. The traditional design disciplines on the left are centred around the product or a technology. Here the designer gains the skills needed to expertly conceive of and give shape to products such as brand identities, interior spaces, buildings, consumer products, etc.

The emerging design practices, on the right, centre around people’s needs or societal needs, and require a different approach in that they need to take longer views and address larger scopes of inquiry. The first author, for example, is currently working on a design research project to explore the unmet needs and dreams of the growing number of people

Table 1. A snapshot in time of traditional and emerging design practices.

The traditional design disciplines focus on the designing of ‘products’ ...	... while the emerging design disciplines focus on designing for a purpose
visual communication design	design for experiencing
interior space design	design for emotion
product design	design for interacting
information design	design for sustainability
architecture	design for serving
planning	design for transforming

who are living with Type 2 Diabetes, in order to identify new opportunities for their future living experiences. The opportunities are open-ended and are not necessarily directed towards product or service manifestations. Another current project involves the use of co-designing tools and thinking in the cultural transformation of people who will be engaged in future healthcare experiences, both from the patient and the healthcare staff side.

Two potential projects also address larger societal needs. One project will look into participatory workplace interventions in acute care nursing with a focus on the psychological and ergonomic challenges involved with obese patients. The other one involves participatory workplace interventions aimed at the workers from family-owned farms in the US Midwest.

The emerging design practices will change what we design, how we design, and who designs. The impact upon the education of designers will be immense. The patterns of change taking place in the transition from a product perspective to the purpose perspective are described more fully in the following sections.

3. The roles in the design process are changing

The move from user-centred design to co-designing is having an impact on the roles of the players in the design process. Let’s consider, for example, the roles of user, researcher and designer (see Figure 3). In a caricature of the classical user-centred design process, the user is a passive object of study, and the researcher brings knowledge from theories and develops more knowledge through observation and interviews. The designer then passively

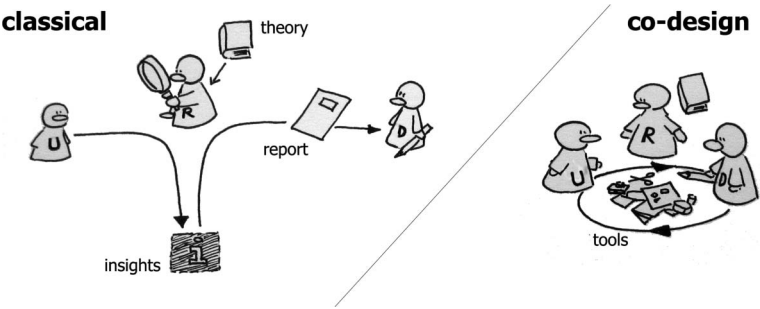


Figure 3. Classical roles of users, researchers, and designers in the design process (on the left) and how they are merging in the co-designing process (on the right).

receives this knowledge in the form of a report and adds an understanding of technology and the creative thinking needed to generate ideas, concepts, etc.

In co-design, on the other hand, the roles get mixed up: the person who will eventually be served through the design process is given the position of ‘expert of his/her experience’, and plays a large role in knowledge development, idea generation and concept development. In generating insights, the researcher supports the ‘expert of his/her experience’ by providing tools for ideation and expression. The designer and the researcher collaborate on the tools for ideation because design skills are very important in the development of the tools. The designer and researcher may, in fact, be the same person. The designer still plays a critical role in giving form to the ideas (more on the changing roles of the key players in the co-design process is discussed in Sleeswijk Visser *et al.* 2005).

3.1. The role of the user: co-designer?

Sometimes ‘users’ can play co-creating roles throughout the design process, i.e. become co-designers, but not always. It depends on level of expertise, passion, and creativity of the ‘user’. All people are creative but not all people become designers. Four levels of creativity can be seen in people’s lives: doing, adapting, making and creating (see Table 2, and Sanders 2006b) These four levels vary in terms of the amount of expertise and interest needed. Expertise, interest/passion, effort, and returns grow with each level.

Table 2. Four levels of creativity.

Level	Type	Motivated by	Purpose	Example
4	Creating	Inspiration	‘express my creativity’	Dreaming up a new dish
3	Making	Asserting my ability or skill	‘make with my own hands’	Cooking with a recipe
2	Adapting	Appropriation	‘make things my own’	Embellishing a ready-made meal
1	Doing	Productivity	‘getting something done’	Organising my herbs and spices

People live simultaneously at all levels of creativity in different parts of their daily lives. For example, they may be at the *creating* level when it comes to cooking but at the *adapting* level when it comes to the use of technology products. People with a high level of passion and knowledge in a certain domain who are invited to participate directly in the design process can certainly become co-designers. For example, this is beginning to take place today with physicians, nurses and other healthcare professionals in the design of new healthcare systems and environments. (Observations based on the first author’s role on planning and architectural design teams.) Sometimes patients and family members become a part of the design team as well.

Fischer (2002) indicates a similar division of levels in viewing people both as consumers and designers, i.e. the same person is and wants to be a consumer in some situations and wants to be a designer in other situations. He proposes a continuum ranging from passive consumer, to active consumer, to end user, to user, to power user, to domain designers, all the way to meta-designer. In relation to the use of products, Fischer advocates the emergence of adaptive design, in which the user could scale his involvement with a product from passive consumption to expert adapting.

Users can become part of the design team as ‘expert of their experiences’ (Sleeswijk Visser *et al.* 2005), but in order for them to take on this role, they must be given appropriate tools for expressing themselves. Over the past decade, research groups within

a number of academic institutions, practitioners in design research consultancies and design research groups within industrial institutes have all explored co-designing tools and techniques and the processes by which they can be applied. The interest in tools and techniques for co-designing are growing rapidly. Figures 4–6 illustrate some of these techniques that come from the authors' most recent personal experiences.

3.2. *The role of the researcher: from translator to facilitator*

In the traditional design process, the researcher served as a translator between the 'users' and the designer. In co-designing, the researcher (who may be a designer) takes on the role



Figure 4. Tools and techniques support the user taking the role of an experienced expert. This photograph shows a presentation technique with a cartoonish TV-frame that can help shy people to express their opinions more readily (see van Rijn and Stappers 2007).



Figure 5. This photograph shows nurses co-creating a concept for ideal workflow on a patient floor. Note that the toolkit components are round, helping them to think in terms of activities, not rooms. This session preceded the one shown below (Sanders 2006c).



Figure 6. This photograph shows nurses co-designing the ideal future patient room using a three-dimensional toolkit for generative prototyping (Sanders 2006c).

of a facilitator. When we acknowledge that different levels of creativity exist, it becomes evident that we need to learn how to offer relevant experiences to facilitate people's expressions of creativity at all levels. This means leading, guiding, and providing scaffolds, as well as clean slates to encourage people at all levels of creativity. It is not always the case that we want to push people beyond their level of interest, passion and creativity. Different approaches to inviting and involving future users into the design development process will be needed for the different levels of creativity. As researchers we will need to learn how to:

- **lead** people who are on the 'doing' level of creativity,
- **guide** those who are at the 'adapting' level,
- **provide scaffolds** that support and serve peoples' need for creative expression at the 'making' level, and
- **offer a clean slate** for those at the 'creating' level.

In addition to bringing people into the design process in the ways most conducive to their ability to participate, researchers will need to bring in applicable domain theories in a way that can be handled by the co-design team. The social psychologist who becomes a design researcher, for example, does not just bring interviewing skills, etc., but can also bring in background knowledge and literature on theories about social interactions, indicating patterns and factors which can guide and/or inspire the design (e.g. Postma and Stappers 2006).

3.3. *The role of the professional designer*

What role is there for professional designers to play if future users are co-creating tangible visions of new products and/or services? Design skills will become even more important in the future as the new landscapes of design emerge. Designers will be in demand as the usefulness of design thinking is acknowledged in mankind's drive to address the challenges of global, systemic issues.

Designers will be needed because they hold highly developed skills that are relevant at larger levels of scope and complexity. By selection and training, most designers are good at visual thinking, conducting creative processes, finding missing information, and being able to make necessary decisions in the absence of complete information. In the near future, designers will find themselves involved not only in the design of stand-alone products but in the design of environments and systems for delivering healthcare, for example. The design of a new community hospital may be completed 8–10 years before the hospital itself is opened. What will the technology be 10 years in the future? Who will be the patients? What will the needs of patients be? Who will be the healthcare workers? How will the transition into the new facility be staged? How will the healthcare workers learn to work in the new facility? As the scope and complexity of design problems increases, we will need the special skills and abilities of designers to help in the way ahead.

We will be using generative design thinking to address change in the future. The use of generative design tools lets one look forward into the possible futures of the people who will be living, working and playing there. The onus is on designers to explore the potential of generative tools and to bring the languages of co-designing into their practice. Designers will be integral to the creation and exploration of new tools and methods for generative design thinking. Designers in the future will make the tools for non-designers to use to express themselves creatively.

Third, designers will need to play a role on the co-designing teams because they provide expert knowledge that the other stakeholders do not have. Designers professionally keep track of existing, new and emerging technologies, and have an overview of production processes and business contexts. This knowledge will still be relevant throughout the design development process.

Fourth, even in the design profession there is considerable specialisation. The skills, knowledge and methods of the interior designer, the interaction designer, the graphic designer, etc., are quite different. These professions will not disappear overnight as ‘users’ become co-designers (Buxton 2005). Expertise within a problem area will remain important even as new design practices emerge (as indicated in Table 1).

4. What does co-creation suggest for changing design practice and education?

Bringing co-creation into design practice will cause a number of changes to occur. It will change how we design, what we design, and who designs. It will also affect the tools and methods that the new teams of co-designers will use.

The landscapes of design and design research will continue to change as design and research blur together. At the front end, design will become synonymous with design research, creating new landscapes of opportunity for designers and researchers. The fuzzy front end will become populated with hybrid design researchers and research designers. Already research is becoming more prominent in the curricula of the quickly growing university-based design programs, and links between, e.g., the social sciences and design are getting stronger. For example, several schools are including participatory design techniques, ethnography and psychology into the curriculum of industrial design engineers (e.g. for the Delft programme, see Stappers *et al.* 2007a, Stappers, Hekkert and Keyson 2007b). In practice we now see industrial designers with many years of experience in product development who are moving into new roles as design researchers.

The design/research blur will be disruptive at first, with arguments going back and forth about who is best suited to do what, which tools and methods belong to whom and how to analyse the data. This is where we are now. As the disruption gains momentum,

however, new disciplines will spin out and people will begin to explore the new design spaces on the emerging landscape. From the blur will come new types of designers and researchers with specialties based more on the purpose of designing as opposed to the products of designing (see Table 1). Or perhaps the spinout from the blur will result in professionals who have special expertise in certain stages of the process, such as in the fuzzy front end.

Co-designing teams will be far more diverse than they are today. Future co-designing will be a close collaboration between all the stakeholders in the design development process together with a variety of professionals having hybrid design/research skills. These team players will vary across many types of culture simultaneously: disciplinary culture, company culture, ethnic culture, worldview, mindset, etc. For example, one of the major challenges in the planning and architectural practices today is the communication gap between the design team, the various levels of 'user groups' and the wide array of specialised consultants to the process. In the future, the new co-design languages that support and facilitate the many varieties of cross-cultural communication will become highly valued.

What is being designed will change. Larger views across space and time will be needed. New tools and methods for design research will be needed to address increasing scope, scale and complexity. The domains of architecture and planning are the last of the traditional design disciplines to become interested in exploring the new design spaces that focus on designing for a purpose. Design for sustainability has been the first of the new design spaces to impact architecture and planning, followed by design for experiencing. The exploration of new design spaces within architecture is happening now primarily in the design of healthcare environments (based on observations from the first author's recent experiences in practice). This is a domain of vast complexity where the introduction of co-designing is being warmly welcomed by many of the healthcare professionals. The opportunity to bring the practice of co-design tools and methods to the design of educational environments and to the corporate workplace is now beginning as well.

4.1. *New domains of collective creativity*

We will see the emergence of new domains of collective creativity that will require new tools and methods for researching and designing. We will need to provide alternative learning experiences and curricula for those who are designing and building scaffolds to support the collective creativity of others. New curricula will be needed for helping people learn how best to function in the generative front end of the product development process. Many questions arise from this challenge. Answers to these questions will require the collective attention of many explorers.

- Is it best for students to learn the traditional design process first and then move to co-designing? Or is it better to start with co-designing? For many people, the mindset for co-designing, i.e. the belief that all people are creative, is difficult to embrace. If we start the education of designers with co-designing, the mindset is less likely to be a problem to overcome.
- When should education about co-creation and co-designing begin? If started in the elementary level, how much more rapidly will the world take shape?
- Who are the new hybrid researcher-designers? Can we learn how to identify people with the most promise? What abilities, skills, mindsets and worldviews will they need?

- What about the learning process for social scientists? Is the learning process for co-designing easier for social scientists (vs. engineers) to learn? What implications does this have for the future?
- What will be the impact of enabling and supporting multiple simultaneous levels of collective creativity? How will that influence design cultures? What influence will that have on the world cultures?
- With everyday people given access to the design process in co-creation, how much will design thinking diffuse into everybody's curriculum for living, learning and working?

In the future, we will be designing in virtual and hybrid domains. We are heading into a world where experience often trumps reality. We can see this now in the rapid proliferation of virtual experiential realms on the internet. The human-centred design research landscape will become a virtual realm, growing and changing in response to the zeitgeist. It will eventually become evident that the design research community does not need to worry about ownership of spaces on the design research landscape since we will be creating new ones. The new landscapes of design and research will be infinite in space and time and continually changing.

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